Title Page

Title： The Association Between Hepatitis C Virus Infection and Renal Cell Cancer, Prostate Cancer, and Bladder Cancer: A Systematic Review and Meta-Analysis

Authors:
Yucheng Ma, yuchengma88@163.com
Zhongli Huang, huangzhongli1234@gmail.com
Zhongyu Jian, 1178590069@qq.com
Xin Wei*. weixinscu@scu.edu.cn

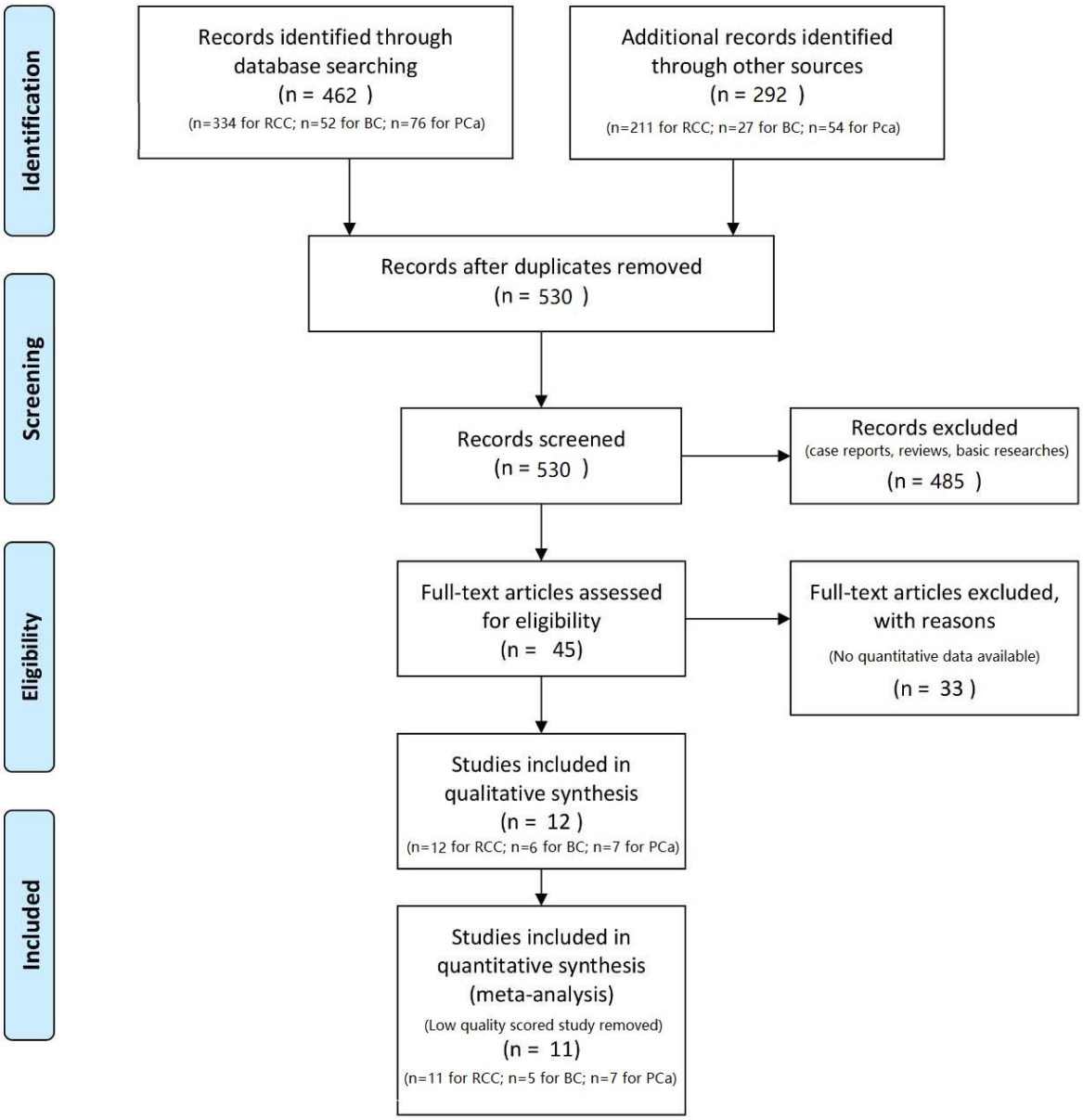
Yucheng Ma, Zhongli Huang and Zhongyu Jian contribute equally

Affiliations of authors: Department of Urology, Institute of Urology（Laboratory of Reconstructive Urology）, West China Hospital, Sichuan University, Chengdu, Sichuan, P.R.C;

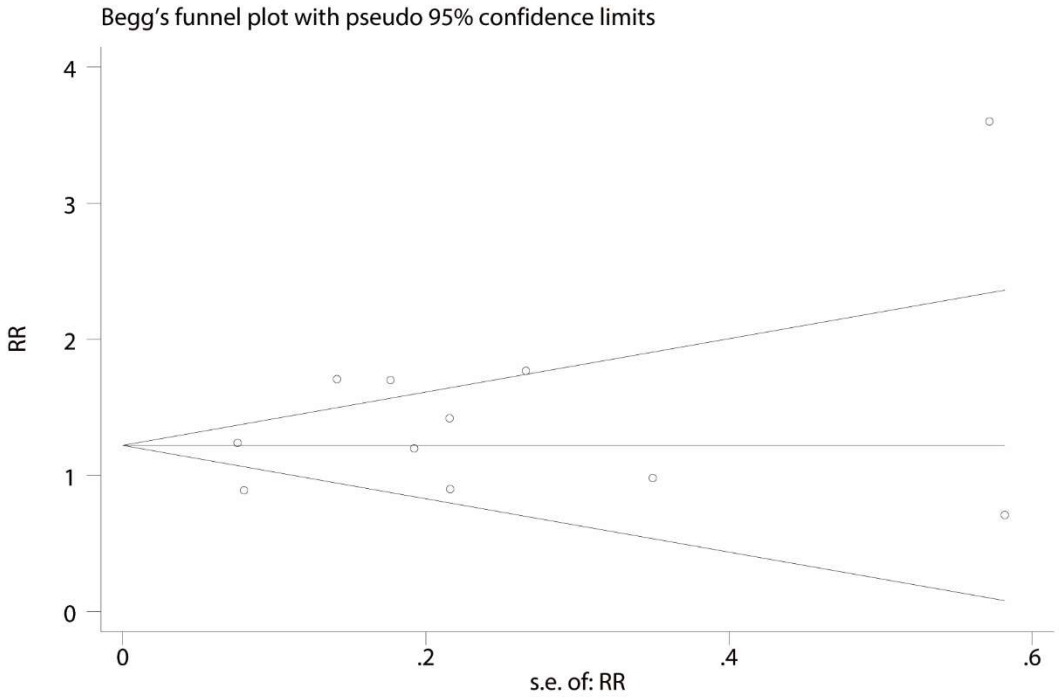
Address: Department of Urology, West China Hospital, Sichuan University, No. 37 Guo Xue Xiang, Chengdu, Sichuan, 610041, P.R. China.

***Correspondence author:** Xin Wei, Department of Urology, Institute of Urology（Laboratory of Reconstructive Urology）, West China Hospital, Sichuan University, Chengdu, 610041, China. Email: weixinscu@scu.edu.cn， Phone: +86 18982081721

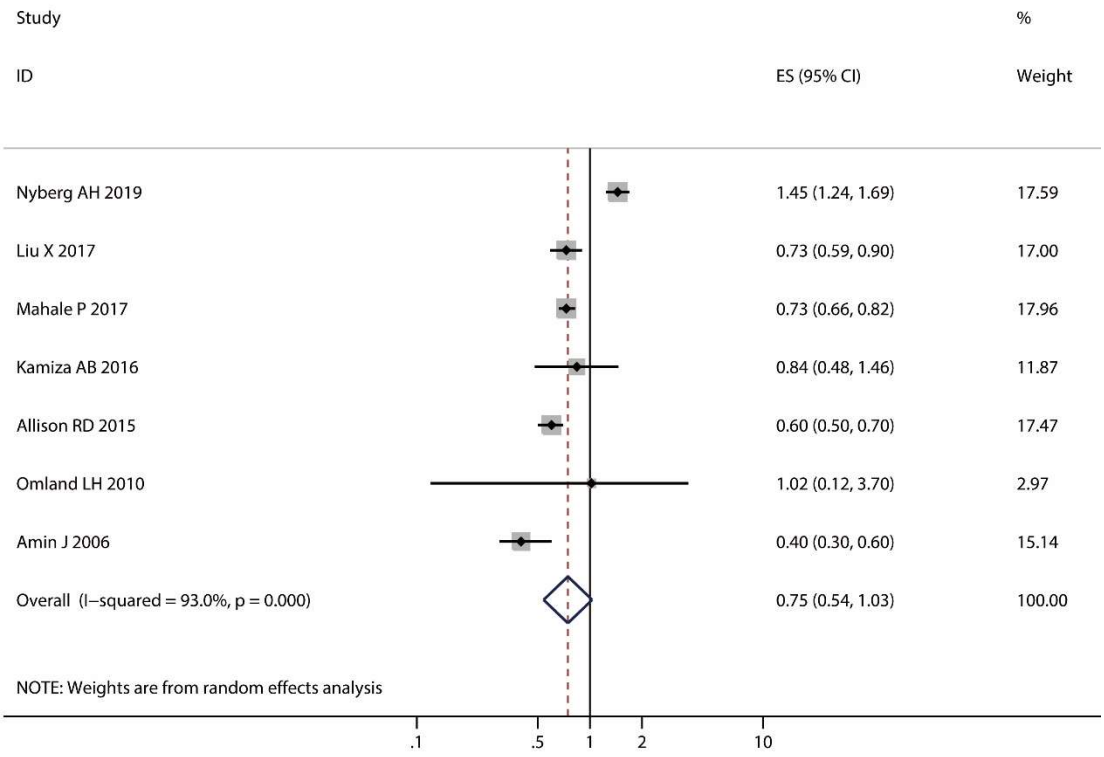
Supplementary Figure S1 Preferred Reporting Items for Systematic Reviews and meta-analysis flowchart.



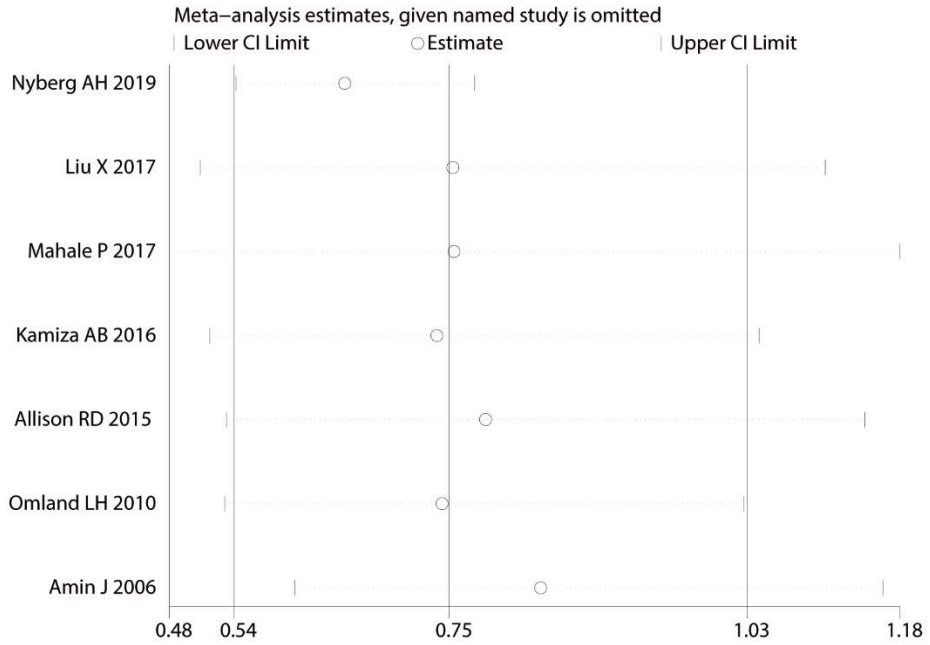
Supplementary Figure S2 Begg test to detect publication bias in meta-analysis investigating the association between hepatitis C virus infection and renal cell cancer.



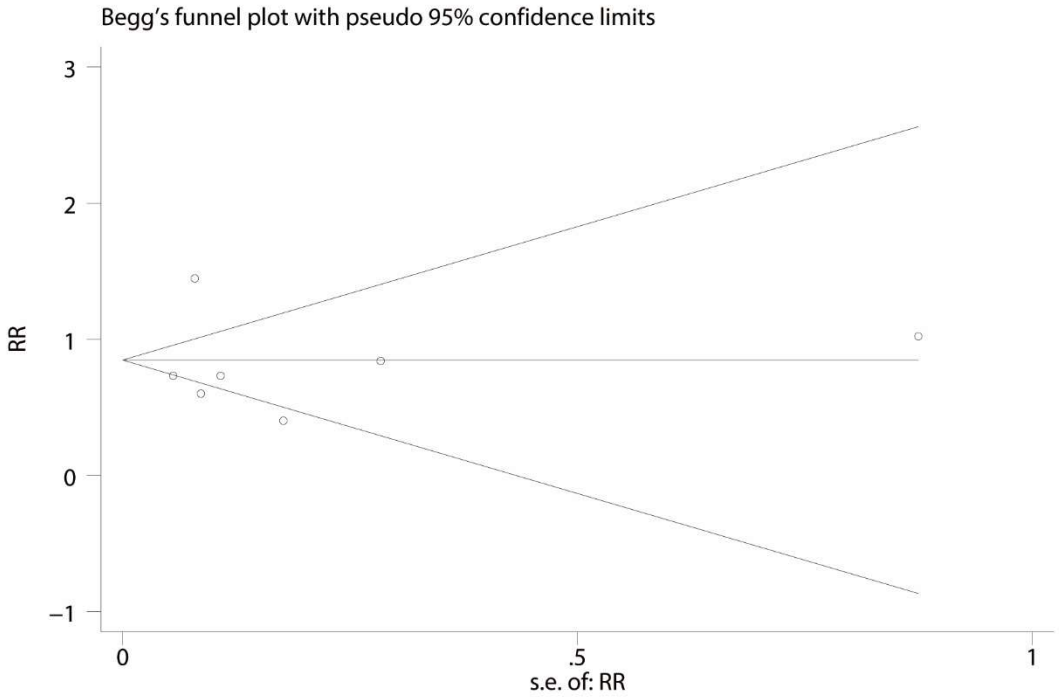
Supplementary Figure S3 Forest plots of relative risk of studies investigating the association between hepatitis C virus infection and prostate cancer. Random effects models were used for the primary meta-analysis.



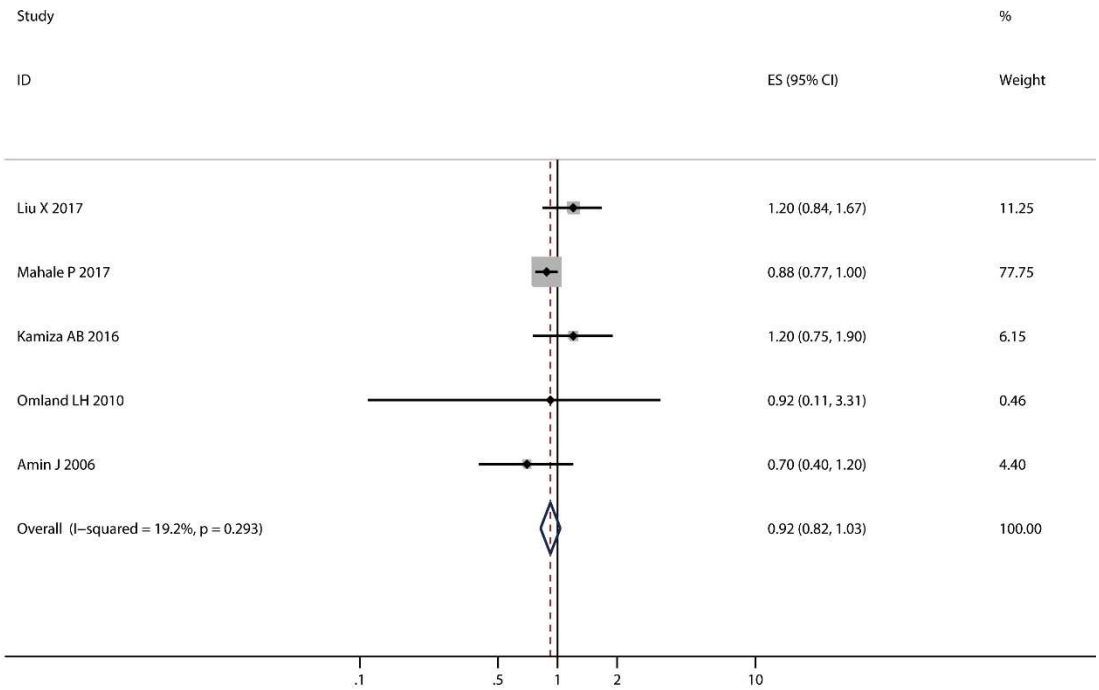
Supplementary Figure S4 Sensitivity analyses by omitting individual study in meta-analysis investigating the association between hepatitis C virus infection and prostate cancer.



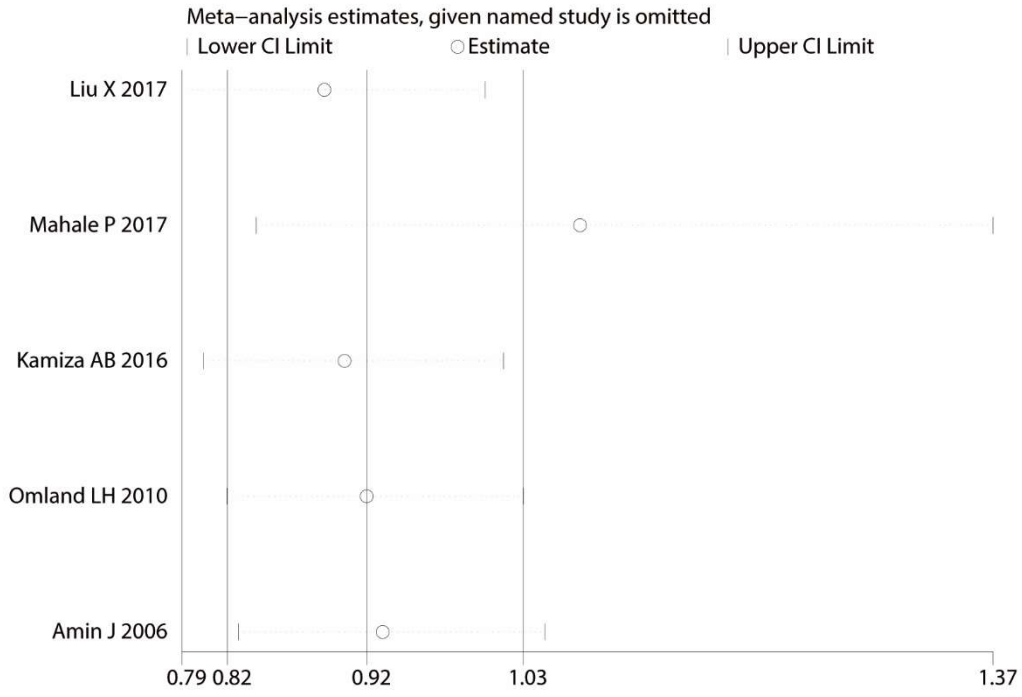
Supplementary Figure S5 Begg test to detect publication bias in meta-analysis investigating the association between hepatitis C virus infection and prostate cancer.



Supplementary Figure S6 Forest plots of relative risk of studies investigating the association between hepatitis C virus infection and bladder cancer. Fixed effects models were used for the primary meta-analysis.



Supplementary Figure S7 Sensitivity analyses by omitting individual study in meta-analysis investigating the association between hepatitis C virus infection and bladder cancer.



Supplementary Figure S8 Begg test to detect publication bias in meta-analysis investigating the association between hepatitis C virus infection and bladder cancer.

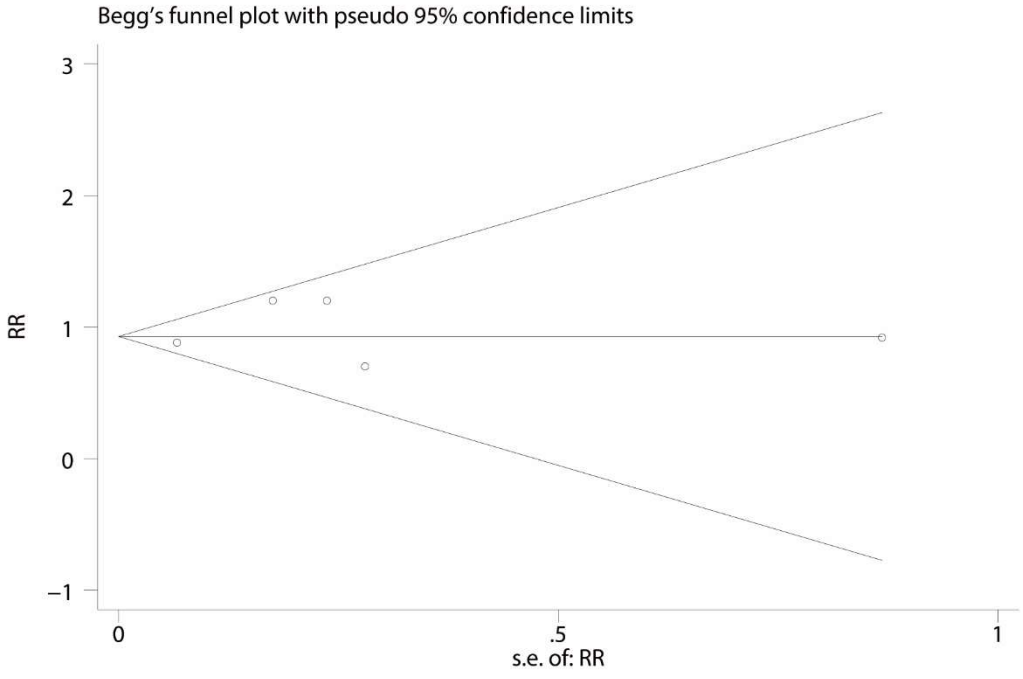


Table S1. Methodological quality of the case-control studies, based on the NOS for assessing the quality of epidemiological studies.

				Selection		**Comparability**		**Exposure**			
Study	Year	Definition of cases	Represent ativeness of cases	Selection of controls	Definition of controls	Important Factor	Additional factor	Ascertain ment	Same method for subjects	Non-respo nse rate*	Totle score
Liu B	2019	1	1	1	0	0	1	0	1	0	5
Lin YS	2017	1	1	1	0	1	1	0	1	1	7
Mahale P	2017	1	1	1	1	1	1	0	1	1	8
Kamiza AB	2016	1	1	1	0	1	0	0	1	1	6

Table S2. Methodological quality of the cohort studies, based on the NOS for assessing the quality of epidemiological studies.

				Selection		**Comparability**		**Outcome**			
Study	Year	Representativeness of exposed cohort	Representativeness of unexposed cohort	Ascertainment of exposure	Outcome was not present at start	Important Factor	Additional factor	Assessment	Exposure Follow-up long enough for outcomes to occur	Adequacy of follow-up	Totle score
Nyberg AH	2019	1	1	1	0	1	1	1	0	1	7
Liu X	2017	1	1	1	1	0	0	0	1	1	6
Allison RD	2015	1	1	1	1	1	1	1	0	1	8
Hofmann JN	2011	1	1	0	0	0	1	1	1	1	6
Omland LH	2010	1	1	1	1	1	0	0	1	1	7
Gordon SC	2010	1	1	1	1	0	1	1	1	1	8
Amin J	2006	1	1	1	1	0	1	1	1	1	8